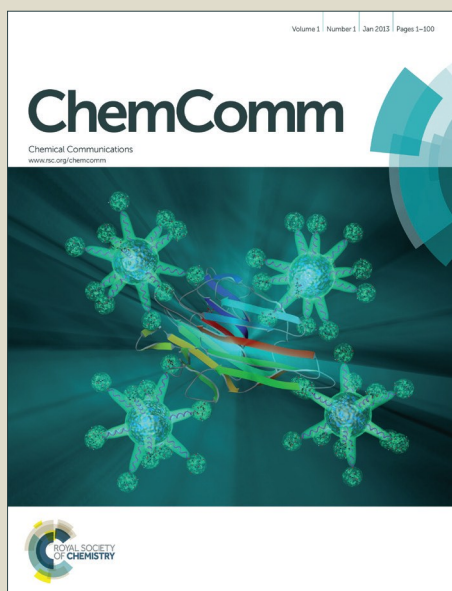


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## COMMUNICATION

# Continuous catalytic upgrading of ethanol to n-butanol over Cu-CeO<sub>2</sub>/AC catalysts

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Received 00th January 20xx,  
Accepted 00th January 20xx

DOI: 10.1039/x0xx00000x

www.rsc.org/

**Cu-CeO<sub>2</sub>/AC catalysts exhibited extraordinary catalytic performance in upgrading of ethanol to n-butanol. To the best of our knowledge, this is the first report that the highest n-butanol yields (21.6% and nearly 20%) could be achieved over heterogeneous catalysts under mild reaction conditions in batch and fixed-bed reactors, respectively. The high catalytic activity, selectivity and stability of these catalysts could be ascribed to the synergy of Cu, CeO<sub>2</sub> and the activated carbon support.**

Upgrading of bio-ethanol to n-butanol via homogeneous or heterogeneous catalytic reactions has currently received great attention both in the scientific and industrial fields, in view of the advantages of n-butanol over bio-ethanol as a sustainable fuel. As more preferable catalysts in industry, heterogeneous catalysts have been comprehensively investigated for n-butanol production from ethanol in last decade. A range of basic zeolites,<sup>1</sup> MgO,<sup>2</sup> Mg-Al metal oxides<sup>3</sup> and hydroxyapatites (HAP)<sup>4</sup> catalyst systems were firstly employed for this reaction. Although high reaction temperature ( $\geq 573$  K) and pressure ( $\geq 6.0$  MPa) were used, the yield of n-butanol was normally lower than 10%. Incorporating metals in catalysts could significantly improve the conversion of ethanol, as a result of the enhanced dehydrogenation of ethanol on metals. For example, Ni/Al<sub>2</sub>O<sub>3</sub> catalyst exhibited up to 80% of selectivity to n-butanol at 25% ethanol conversion after reaction of 72 h at 523 K and 7 MPa (Ar) in a batch reactor.<sup>5</sup> However, due to the strong cracking capability of C-C bonds of metal Ni, considerable amount of gaseous carbon products such as CO, CO<sub>2</sub>, CH<sub>4</sub>, were also detected. Similar catalytic performance was reported in a fixed-bed reactor over basic aluminum oxide supported Cu, Ni and Co catalysts for continuous catalytic upgrading of ethanol to n-butanol at 513 K and 7 MPa.<sup>6</sup> More recently, Poliakov and his co-workers have exploited a high surface area CeO<sub>2</sub> (HSA-CeO<sub>2</sub>)

supported Cu catalyst for upgrading ethanol to n-butanol in the presence of supercritical CO<sub>2</sub> in a fixed-bed reactor,<sup>7</sup> which exhibited 45% selectivity to n-butanol and up to 30% yield of n-butanol. Though the yield of n-butanol is the highest ever reported on heterogeneous catalysts, the use of supercritical system with reaction pressure higher than 10 MPa requires a specific reaction equipment and complicated process. Therefore, the development of new heterogeneous catalysts that have superior activity and selectivity for this reaction under mild conditions while can be easily synthesized is highly desired for future industrial n-butanol production.

Active carbon (AC) is widely used as catalyst itself or support for precious metals such as Pd, Pt and Ru,<sup>8</sup> due to its unique properties such as extremely high specific surface area, controllable porosity and surface chemistry, and high stability at various pH values. On the other hand, Cu-based catalysts, especially Al<sub>2</sub>O<sub>3</sub> and SiO<sub>2</sub> supported Cu catalysts are well known as very active heterogeneous catalysts for various reactions, such as steam reforming of methanol, CO oxidation, hydrogenation and dehydrogenation reactions.<sup>9</sup> However, AC supported copper catalysts are less applied in catalytic reactions, because of the liable sintering of Cu nanoparticles engendered by the weak metal-support interaction.<sup>10</sup> In the present work, we first reported that Cu-CeO<sub>2</sub>/AC catalysts prepared by a simple impregnation method demonstrated superior catalytic performance for ethanol upgrading to n-butanol.

The catalytic performance of Cu-CeO<sub>2</sub>/AC catalysts with different Cu/Ce molar ratios, Cu/AC, 3Cu1Ce/SiO<sub>2</sub> and 3Cu1Ce/Al<sub>2</sub>O<sub>3</sub> is displayed in Table 1. The Cu/AC catalyst showed only 1.2% of ethanol conversion and 12.1% of selectivity to n-butanol whereas 55.3% of selectivity to 1,1-diethoxyethane in a fixed-bed reactor. The addition of Ce dramatically improved the catalytic performance of Cu-based catalysts for ethanol upgrading to n-butanol. Cu-CeO<sub>2</sub>/AC catalysts, in spite of different Cu/Ce molar ratios, all exhibited higher ethanol conversions (44.0-46.2%), selectivities to n-butanol (36.8-42.4%) and n-butanol yields (16.3-19.3%).

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<sup>†</sup> Electronic supplementary information (ESI) available: Details of experimental methods, supporting figures, tables and schemes.

**Table 1.** Catalytic performance of Cu-CeO<sub>2</sub>/AC catalysts with the different Cu/Ce molar ratios<sup>[a]</sup>

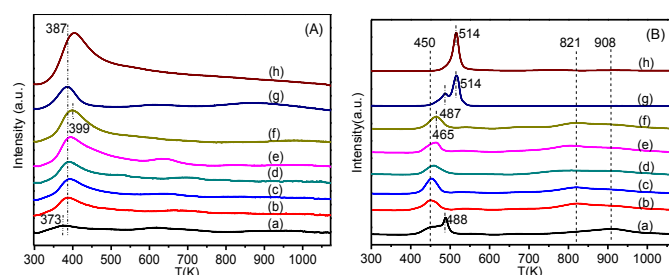
| Catalyst                              | Conversion (%) | Selectivity(%) |               |               |         |                            |                       | Butanol Yield (%) |
|---------------------------------------|----------------|----------------|---------------|---------------|---------|----------------------------|-----------------------|-------------------|
|                                       |                | Acetaldehyde   | Butyraldehyde | Ethyl Acetate | Butanol | C6 products <sup>[b]</sup> | Others <sup>[c]</sup> |                   |
| Cu/AC                                 | 1.2            | 10.5           | 0.5           | 0.5           | 12.1    | -                          | 76.4                  | 0.1               |
| 5Cu1Ce/AC                             | 46.2           | 6.8            | 8.2           | 10.3          | 41.2    | 20.6                       | 12.9                  | 19.0              |
| 4Cu1Ce/AC                             | 45.6           | 7.8            | 7.8           | 8.5           | 42.4    | 20.3                       | 12.8                  | 19.3              |
| 3Cu1Ce/AC                             | 46.2           | 7.2            | 7.8           | 10.9          | 41.3    | 19.7                       | 13.0                  | 19.1              |
| 2Cu1Ce/AC                             | 46.3           | 6.6            | 7.5           | 12.5          | 39.3    | 19.4                       | 14.6                  | 18.2              |
| 1Cu1Ce/AC                             | 44.0           | 5.6            | 6.7           | 17.4          | 36.8    | 8.7                        | 24.8                  | 16.3              |
| 3Cu1Ce/SiO <sub>2</sub>               | 23.3           | 17.1           | 17.0          | 45.4          | 10.2    | 2.1                        | 8.1                   | 2.4               |
| 3Cu1Ce/Al <sub>2</sub> O <sub>3</sub> | 46.9           | 4.4            | 16.7          | 44.7          | 12.1    | 5.3                        | 16.8                  | 5.7               |
| 3Cu1Ce/AC <sup>[d]</sup>              | 39.1           | 1.1            | 1.8           | 6.3           | 55.2    | 20.3                       | 15.2                  | 21.6              |

[a] Conversion, selectivity and yield are obtained at steady-state; reaction conditions: catalyst, 1.0 g; 523 K, 2 MPa (N<sub>2</sub>), LHSV=4 ml/(h·g·cat), N<sub>2</sub>/ethanol(v/v)=500:1; [b] C6 products include 2-ethylbutyraldehyde, hexaldehyde, 2-ethylbutanol, and 1-hexanol; [c] Other products include 1-ethoxyethane, 1,1-diethoxyethane, butyl acetate, etc; specially for Cu/AC, 2.6% 1-ethoxyethane, 55.3% 1,1-diethoxyethane; for Cu-CeO<sub>2</sub>/AC catalysts with Cu/Ce molar ratio of 1:1, 5.0% 1-ethoxyethane, 9.3% 1,1-diethoxyethane. [d] Reaction was carried out in a batch reactor with volume of 250 mL. Reaction conditions: catalyst, 3.3 g (<120 mesh); 80 g absolute ethanol; 523 K, 0.1 MPa (N<sub>2</sub>); reaction time 48 h.

In contrast, Cu-CeO<sub>2</sub> supported on SiO<sub>2</sub> or Al<sub>2</sub>O<sub>3</sub> rather than AC had a lower ethanol conversion and selectivity to n-butanol, but much higher selectivity to ethyl acetate. Additionally, 39.1% of ethanol conversion, 55.2% of selectivity to n-butanol and up to 21.6% of n-butanol yield were also achieved over the 3Cu1CeO<sub>2</sub>/AC catalyst after reaction of 48 h at 523 K and 0.1 MPa (N<sub>2</sub>) in a batch reactor. Comparing with other heterogeneous catalyst systems reported in literature for catalytic production of n-butanol from ethanol, such as Mg-Al mixed oxide catalysts,<sup>3</sup> HAP catalysts,<sup>4</sup> Ni/Al<sub>2</sub>O<sub>3</sub> catalysts<sup>5,6</sup> and Cu/CeO<sub>2</sub> catalysts,<sup>7</sup> the Cu-CeO<sub>2</sub>/AC catalysts exhibited much higher ethanol conversions and selectivities to n-butanol even under much milder reaction conditions both in batch and fixed-bed reactors. As shown in Fig. S1 (ESI<sup>†</sup>), these Cu-CeO<sub>2</sub>/AC catalysts also possessed good stability in the 12 h evaluation on stream, showing a promising prospect for industrial applications. In addition, possible gaseous products (via ethanol cracking, decarbonylation of acetaldehyde, etc.) such as CH<sub>4</sub>, CO and CO<sub>2</sub> besides H<sub>2</sub> were not detected in the outlet gas, confirming the cleavage of C-C bonds is not favorable over metal Cu.

In order to shed light on the mechanism of the superior catalytic performance of Cu-CeO<sub>2</sub>/AC catalysts, XRD, TEM and N<sub>2</sub> adsorption analysis was employed for reduced catalysts before and after reaction (Fig. S2 and Table S1, ESI<sup>†</sup>). As shown in Table S1, the average diameters of Cu nanoparticles calculated according to Scherrer formula for Cu/AC and Cu-CeO<sub>2</sub>/AC catalysts before reaction are all ranged from 4 to 6 nm. After 12h of reaction, the Cu average crystalline diameter of Cu/AC catalyst dramatically increased from 4.3 nm to 22.0 nm, suggesting the growth of Cu metals during the reaction, while these of Cu-CeO<sub>2</sub>/AC catalysts are still less than 10 nm. These

results show that the addition of Ce restrains the sintering and growth of Cu nanoparticles on the surface of AC support to a great extent, and thus the Cu metals still maintain high dispersion over the Cu-CeO<sub>2</sub>/AC catalysts. As for the CeO<sub>2</sub> crystallites over Cu-CeO<sub>2</sub>/AC catalysts, the average crystalline diameters of CeO<sub>2</sub> for fresh and spent Cu-CeO<sub>2</sub>/AC catalysts are all less than 5 nm (Table S1), suggesting CeO<sub>2</sub> species are also highly dispersive over the Cu-CeO<sub>2</sub>/AC catalysts. As shown in Table S1, the specific surface area of Cu/AC catalyst is higher than those of Cu-CeO<sub>2</sub>/AC catalysts, but the rapid growth of Cu metals was observed over Cu/AC catalyst after reaction. It thus implies that the high dispersion of Cu and CeO<sub>2</sub> over Cu-CeO<sub>2</sub>/AC catalysts after reaction should be the combinational result of the large specific surface area of AC and the strong interaction between Cu and CeO<sub>2</sub>.



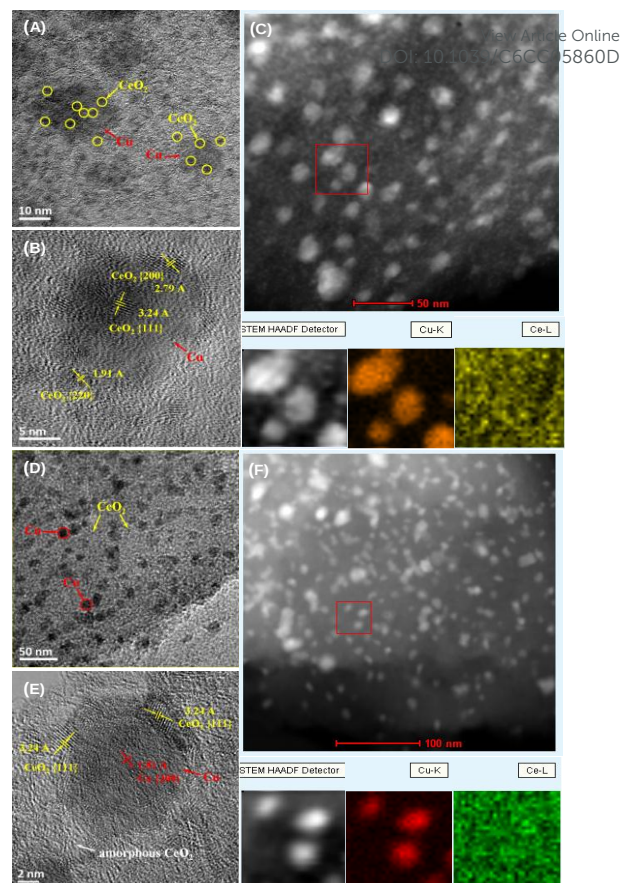
**Fig. 1.** (A) CO<sub>2</sub>-TPD profiles of reduced catalysts and (B) TPR profiles of the calcined catalysts: (a) Cu/AC; (b) 5Cu1Ce/AC; (c) 4Cu1Ce/AC; (d) 3Cu1Ce/AC; (e) 2Cu1Ce/AC; (f) 1Cu1Ce/AC; (g) 3Cu1Ce/SiO<sub>2</sub>; (h) 3Cu1Ce/Al<sub>2</sub>O<sub>3</sub>

Fig. 1A depicts the CO<sub>2</sub>-TPD of the reduced Cu-based catalysts. For the Cu/AC catalyst, a weak CO<sub>2</sub> desorption peak at 373 K can be observed in Fig. 1A(a), which can be attributed to the CO<sub>2</sub> desorption from basic sites on the AC support (Fig. S3,

ESI<sup>†</sup>). Another weak and broad peak appearing at higher temperature from 545 K to 765 K can be ascribed to the desorption of CO<sub>2</sub> from Cu metals.<sup>11</sup> Over Cu-CeO<sub>2</sub>/AC catalysts, as shown in Fig. 1A(b)-(f), a large CO<sub>2</sub> desorption peak (300-500 K) can be observed. This desorption peak should be the result of CO<sub>2</sub> desorption from highly dispersed CeO<sub>2</sub> species as demonstrated in Fig. S4 (ESI<sup>†</sup>). The intensity of this CO<sub>2</sub> desorption peak significantly increases with the content of Ce, implying the addition of Ce introduced basic sites onto the Cu-based catalysts. Meanwhile, this peak also shifts to higher desorption temperature with the increase in Ce loading, suggesting a stronger CO<sub>2</sub> interaction with the catalysts. This enhanced basicity of the catalysts is likely due to the incorporation of Cu in the crystal lattice of CeO<sub>2</sub> as proved by subsequent TEM analysis, which raises the basicity of surface oxygen on CeO<sub>2</sub>.<sup>12</sup> The contribution of highly dispersed CeO<sub>2</sub> species to the basicity of catalysts can be also observed over 3Cu1Ce/SiO<sub>2</sub> and 3Cu1Ce/Al<sub>2</sub>O<sub>3</sub> catalysts, as shown from Fig. 1A(g)-(h).

Fig. 1(B) shows the TPR profiles of various calcined catalysts. There are three H<sub>2</sub> consumption peaks in the TPR profile of Cu/AC catalyst (Fig. 1B(a)). The high temperature peak centred at 908 K can be ascribed to the reduction of oxygen-containing groups on the surface of AC support, since the AC itself has the same H<sub>2</sub> consumption peak in the similar temperature range (Fig. S5, ESI<sup>†</sup>). The peak at 450 K is assigned to the reduction of CuO with small particle size, while the peak at 488 K is ascribed to the reduction of CuO with larger particle size, wherein the presence of multiple H<sub>2</sub> consumption peaks of CuO species reflects heterogeneous size distribution of Cu nanoparticles. As shown in Fig. 1B(b)-(f), the calcined Cu-CeO<sub>2</sub>/AC catalysts exhibit similar reduction behaviors with two reduction peaks centered at 450-465 K and 821 K, respectively. The low temperature peak is ascribed to the reduction of CuO, while the high temperature peak is assigned to the reduction of CeO<sub>2</sub> (Fig. S6, ESI<sup>†</sup>). The reduction peak of CuO slightly shifts to higher temperature relative to that of Cu/AC catalyst, meaning a strong interaction existing between Cu and CeO<sub>2</sub>.<sup>13</sup> Moreover, the appearance of single reduction peak of CuO species demonstrates that CuO nanoparticles are more uniformly dispersed on these catalysts.

The calcined Cu-CeO<sub>2</sub>/SiO<sub>2</sub> catalyst exhibits a similar TPR profile as that of the Cu/AC catalyst at the low temperature range (Fig. 1B(g)). The shoulder peak at 487 K and main peak at temperature of 514 K are also assigned to CuO species with small particle size and large particle size, respectively. The up-shift of the reduction temperature of CuO species compared with that of Cu/AC catalyst implies the stronger metal-support interaction on Cu-CeO<sub>2</sub>/SiO<sub>2</sub> catalyst. However, the calcined Cu-CeO<sub>2</sub>/Al<sub>2</sub>O<sub>3</sub> catalyst displays only one H<sub>2</sub> consumption peak at 514 K (Fig. 1B(h)), which is ascribed to the reduction of CuO species highly dispersed on the surface of Al<sub>2</sub>O<sub>3</sub> support, as revealed in Fig. S2 and Table S1 (ESI<sup>†</sup>). In addition, there is no reduction peak of CeO<sub>2</sub> in the TPR profile of Cu-CeO<sub>2</sub>/Al<sub>2</sub>O<sub>3</sub> or Cu-CeO<sub>2</sub>/SiO<sub>2</sub> catalyst, probably due to the much stronger interaction between CeO<sub>2</sub> and SiO<sub>2</sub> or Al<sub>2</sub>O<sub>3</sub> than activated carbon.



**Fig. 2.** TEM images and corresponding EDX mapping of (A), (B), (C) 5Cu1Ce/AC and (D), (E), (F) 3Cu1Ce/AC catalysts after reaction.

Fig. 2 exhibits the TEM images of two typical Cu-CeO<sub>2</sub>/AC catalysts, 5Cu1Ce/AC and 3Cu1Ce/AC, after reaction. As shown in Fig. 2(A) and (B), the average crystalline size of Cu nanoparticles for the 5Cu1Ce/AC is 10.4 nm, while CeO<sub>2</sub> nanoparticles are highly dispersive with the average size of 1.8 nm. It is especially notable that each of Cu nanoparticles is adhered by several small CeO<sub>2</sub> nanoparticles and then a special structure in which Cu nanoparticles are partially covered by small CeO<sub>2</sub> nanoparticles is formed. As mentioned above, there is a strong metal-support interaction between Cu metals and CeO<sub>2</sub> support. Here, TEM analysis further indicates an incorporation of Cu in the crystal lattice of CeO<sub>2</sub>, as reflected by an expansion of the interplanar spacing of {111} and {200} crystal faces of CeO<sub>2</sub> from 3.12 to 3.24 and 2.71 to 2.79, respectively (Fig. 2(B)). This special structure is thus responsible for the restriction of sintering and growth of Cu nanoparticles and subsequent improved stability of AC-supported Cu-based catalysts. As is shown in Fig. 2(D) and (E), the average crystalline size of Cu nanoparticles for the 3Cu1Ce/AC is 9.1 nm, while Ce species present in a crystal or amorphous form closely surround Cu nanoparticles not only at the perimeter of Cu nanoparticles on the surface of AC support but also on the surface of Cu nanoparticles. This special structure in which Cu nanoparticles are isolated by these Ce species further stabilized the Cu nanoparticles over the Cu-CeO<sub>2</sub>/AC catalysts. Energy Dispersive X-ray (EDX) mapping analysis shows that Cu-K signals can only be detected in the selected bright regions while Ce-L signals are

uniformly spread over the whole area (Fig. 2(C) and (F)). These results indicate a close contact of Cu nanoparticles with the uniformly distributed Ce species on the surface of Cu-CeO<sub>2</sub>/AC catalysts, which is consistent with the bright-field TEM observations at high resolution. Therefore, the addition of Ce species stabilizes Cu nanoparticles on the surface of activated carbon by partially covering Cu nanoparticles (in the case of low Ce loading) and further isolating Cu nanoparticles (in the case of high Ce loading).

The product distribution (Table 1) over Cu-CeO<sub>2</sub>/AC catalysts indicates n-butanol is produced from ethanol via a sequence of steps known as Guerbet reaction. In this cascade process (Scheme S1),<sup>14</sup> ethanol experiences dehydrogenation to form an acetaldehyde, followed by aldol condensation before the resultant product is dehydrated and subsequently rehydrogenated to produce n-butanol, which may undergo further coupling to give higher carbon number alcohols. Kinetic analysis (Figs. S9-15 and Table S4-8, ESI<sup>†</sup>) reveals the apparent activation energy for ethanol dehydrogenation, aldol condensation and rehydrogenation is 32.5, 70.1, 12.4 kJ/mol, respectively, indicating aldol condensation is the decisive step in these tandem steps for n-butanol production. It should be noting all the apparent activation energies calculated here are much lower than these reported over hydroxyapatite.<sup>15</sup> Because dehydrogenation and subsequent rehydrogenation on Cu-CeO<sub>2</sub>/AC catalysts is catalyzed by Cu metals (CeO<sub>2</sub> is relatively inactive for dehydrogenation and Meerwein-Ponndorf-Verley (MPV) reaction, Scheme S2, ESI<sup>†</sup>) and aldol condensation is favorable over basic CeO<sub>2</sub> sites, the highly dispersed Cu metals are beneficial for dehydrogenation and rehydrogenation while the presence of highly distributed CeO<sub>2</sub> species provides sufficient basic sites for the aldol condensation of acetaldehyde, which explains the high activity of Cu-CeO<sub>2</sub>/AC catalysts. Furthermore, it has been reported that Cu<sup>+</sup> species are catalytic active for the formation of ethyl acetate.<sup>16</sup> The higher reduction temperature of CuO over the Cu-CeO<sub>2</sub>/SiO<sub>2</sub> or Cu-CeO<sub>2</sub>/Al<sub>2</sub>O<sub>3</sub> catalyst than those over the Cu-CeO<sub>2</sub>/AC catalysts (Fig. 1B) suggests that there are more Cu<sup>+</sup> species on the former two catalysts after reduction, which is considered responsible for the high selectivity to ethyl acetate. Therefore, the high selectivity to n-butanol instead of ethyl acetate over Cu-CeO<sub>2</sub>/AC catalysts could be related to the synergy of Cu, CeO<sub>2</sub> and the activated carbon support in catalysis.

In summary, highly efficient Cu-CeO<sub>2</sub>/AC catalysts prepared by a wetness impregnation method were used for continuous catalytic upgrading of ethanol to n-butanol. The Cu-CeO<sub>2</sub>/AC catalysts exhibited average 40% of selectivity towards n-butanol and nearly 20% of n-butanol yield at lower temperature (523 K) and pressure (2 MPa N<sub>2</sub>). The high dispersion of Cu metals is favorable for high ethanol conversion, while the presence of highly dispersive Ce species as basic sites is beneficial for the aldol condensation of acetaldehyde, which drives the reaction equilibrium towards n-butanol production. Additionally, the stability of Cu-CeO<sub>2</sub>/AC catalysts can be improved due to the interaction between Cu metals and these Ce species. The yield of n-butanol is also influenced by the nature of the support, so that higher ethyl acetate yield and subsequent lower n-butanol

yield are observed over catalysts containing SiO<sub>2</sub> or Al<sub>2</sub>O<sub>3</sub>. Therefore, the high catalytic activity, selectivity towards n-butanol and n-butanol yield could be ascribed to the synergy of Cu, CeO<sub>2</sub> and the activated carbon support in upgrading ethanol to n-butanol.

This work was supported financially by the National Natural Science Foundation of China (NSFC Grant No. 21303163), the Natural Science Foundation of Zhejiang Province (LY13B030006, LY17B060006), the Qianjiang Talent Project in Zhejiang Province (QJD1302011) and the Scientific Research Fund of Zhejiang Provincial Education Department (Y201328681).

## Notes and references

- 1 Yang, Z. J. Meng, *Catal.* **1993**, *142*, 37.
- 2 a) A. S. Ndou, N. Plint, N. J. Coville, *Appl. Catal. A* **2003**, *251*, 337; b) T. W. Birky, J. T. Kozlowski, R. J. Davis, *J. Catal.* **2013**, *298*, 130.
- 3 a) D. L. Carvalh, M. T. Rodrigues, L. E. P. Borges, L. G. Appel, *Appl. Catal.* **2012**, *415-416*, 96; b) I. C. Marcu, D. Tichit, F. Fajula, N. Tauchoux, *Catal. Today* **2009**, *147*, 231; c) I. C. Marcu, D. Tichit, F. Fajula, N. Tauchoux, *Catal. Lett.* **2013**, *143*, 23.
- 4 a) T. Tsuchida, S. Sakuma, T. Takeguchi, W. Ueda, *Ind. Eng. Chem. Res.* **2006**, *45*, 8634; b) T. Tsuchida, S. Sakuma, T. Takeguchi, W. Ueda, T. Yoshioka, *J. Catal.* **2008**, *259*, 183; c) S. OgO, A. Onda, K. Yanagisawa, *Appl. Catal. A* **2011**, *402*, 188; d) S. Hanspal, Z. D. Young, H. Shou, R. J. Davis, *ACS Catal.* **2015**, *5*, 1737.
- 5 T. Riittonen, E. Toukonitty, D. K. Madnani, A. R. Leino, K. Kordas, *Catalysts* **2012**, *2*, 68.
- 6 T. Riittonen, K. Eränen, P. Mäki-Arvela, A. Shchukarev, A. R. Rautio, K. Kordas, N. Kumar, T. Salmi, J. P. Mikkola, *Renewable Energy* **2015**, *74*, 369.
- 7 J. H. Earley, R. A. Bourne, M. J. Watson, M. Poliakoff, *Green Chem.* **2015**, *17*, 3018.
- 8 a) C. W. Xu, P. K. Shen, Y. L. Liu, *J. Power Source* **2007**, *164*, 524; b) Y. M. Zhang, H. X. Jiang, Y. H. Wang, M. H. Zhang, *Ind. Eng. Chem. Res.* **2014**, *53*, 6380.
- 9 M. B. Gawande, A. Goswami, F. X. Felpin, T. Asefa, X. Huang, R. Silva, X. Zou, R. Zboril, R. S. Varma, *Chem. Rev.*, **2016**, *116*, 3722.
- 10 P. Wynblatt, N. A. Gjostein, *Prog. Solid State Chem.* **1975**, *9*, 21.
- 11 V. S. Guggilla, V. R. R. Pendyala, S. S. Chakravartula, V. R. C. Komandur, *J. Phys. Chem. B* **2006**, *110*, 13881.
- 12 a) S. I. Ilgaz, S. Hyuntae, S. Deepika, T. M. Jeffrey, S. O. Umit, *ACS Catal.* **2014**, *4*, 585; b) M. J. Li, U. Tumuluri, Z. L. Wu, S. Dai, *ChemSusChem* **2015**, *8*, 3651.
- 13 J. A. Farmer, C. T. Campbell, *Science* **2010**, *329*, 933.
- 14 a) J. Scalbert, T. S. Frederic, J. Roland, M. Didier, M. Frederic, *J. Catal.* **2014**, *311*, 28; b) S. Shylesh, D. Kim, C. R. Ho, G. R. Johnson, J. Wu, A. T. Bell, *ChemSusChem* **2015**, *8*, 3959; c) T. Moteki, D. W. Flaherty, *ACS Catal.* **2016**, *6*, 4170.
- 15 C. R. Ho, S. Shylesh, A. T. Bell, *ACS Catal.* **2016**, *6*, 939.
- 16 A. G. Sato, D. P. Volanti, D. M. Meira, S. Damyanova, E. Longo, J. M. C. Bueno, *J. Catal.* **2013**, *307*, 1.